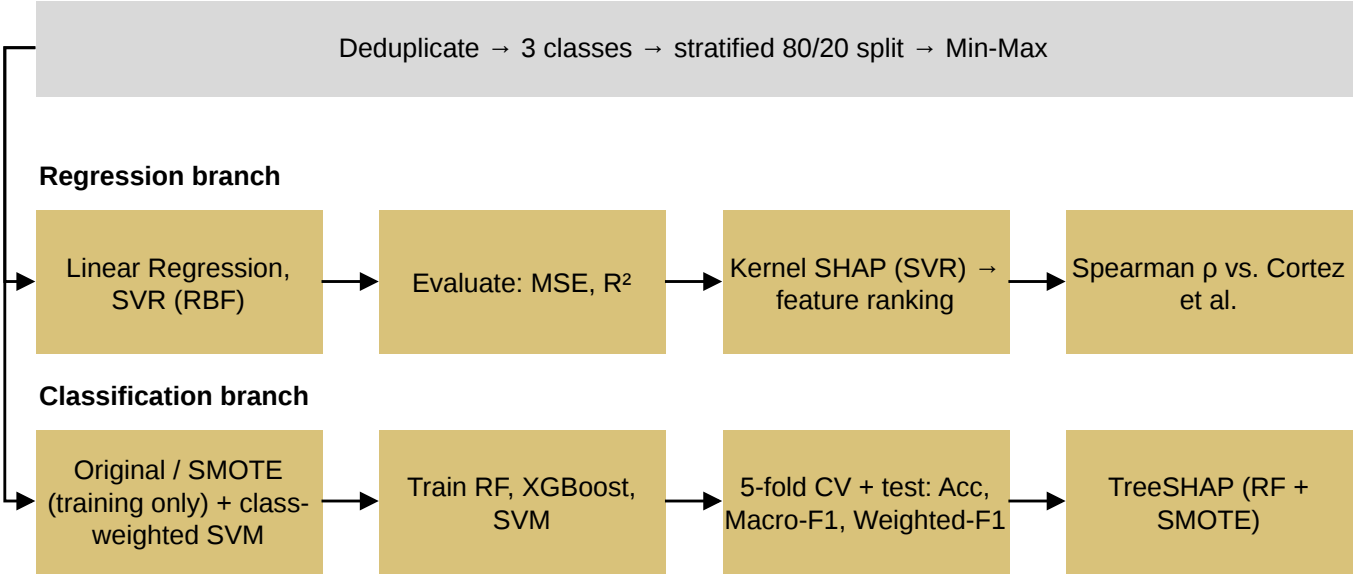
Deduplicate → 3 classes → stratified 80/20 split → Min-Max
Regression branch
Linear Regression, SVR (RBF)
Evaluate: MSE, R²
Kernel SHAP (SVR) → feature ranking
Spearman ρ vs. Cortez et al.
Classification branch
Original / SMOTE (training only) + class-weighted SVM
Train RF, XGBoost, SVM
5-fold CV + test: Acc, Macro-F1, Weighted-F1
TreeSHAP (RF + SMOTE)